

LAPS: A Lightweight Privilege-Allocation Prompting Framework for Source Localization

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Abstract

The widespread use of social-media graphs has provided a convenient channel for rumor propagation. Rapid localization of rumor sources is therefore crucial for mitigating diffusion and enabling punitive countermeasures. Source Localization (SL) aims to identify the origin nodes given partial infection observations. Although deep-learning-based SL approaches outperform traditional estimators, three fundamental limitations remain: (i) *Model Complexity*—existing methods enrich node embeddings with cascades of auxiliary features, yielding high-capacity but excessively complex representations, leading to an exponential increase in the number of model parameters; (ii) *Annotation gap*—to overcome the scarcity of real-world misinformation cascades, current pipelines repeatedly simulate diffusion from a fixed seed, eroding robustness on true, few-shot outbreaks; and (iii) *Computational bottleneck*—full-model retraining or recurrent cascade simulation is required for every new task, which disqualifies the solutions from real-time deployment. Inspired by the success of prompt learning in NLP and graph learning, we propose LAPS, a Lightweight privilege-Allocation Prompting framework for Source localization. LAPS first trims parameter explosion and data scarcity by pre-training a graph-level source region classifier on adaptive subgraphs with source-prior diffusion data. It then enables few-shot SL via a privilege-allocation prompt module that updates $< 1\%$ of all the parameters, avoiding model retraining to facilitate efficiency. Extensive experiments on five real-world networks demonstrate the effectiveness and efficiency of our prompt-based framework on few-shot source localization task.

CCS Concepts

• Information systems → Social networks.

Keywords

Source Localization, Pre-training, Prompt Learning

ACM Reference Format:

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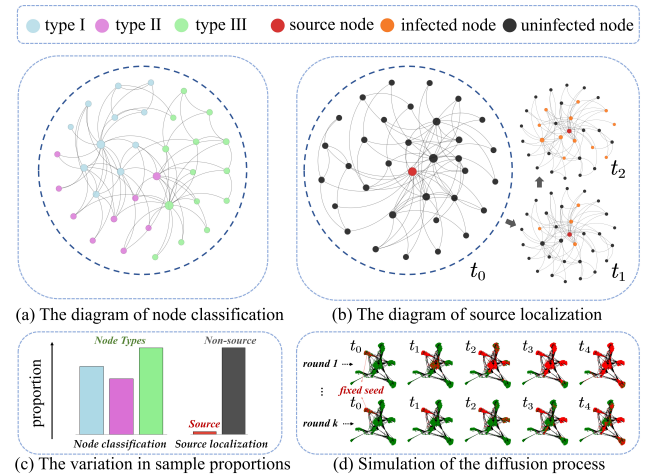


Figure 1: The distinction between node classification and source localization. (a) The diagram of node classification. (b) The diagram of source localization. (c) The sample proportions differ drastically between node classification and source localization. (d) Existing methods predominantly rely on repeatedly acquiring historical diffusion data from fixed sources, where k sets of diffusion data are repeatedly generated based on the same source node.

1 Introduction

In the digital era, social media networks, structured as graph-theoretical models with users as nodes and interactions as edges, have become central to information dissemination, epitomizing the interconnectedness of modern society[35]. However, this connectivity also exposes these networks to significant cybersecurity risks[27], particularly from phishing attacks and malware propagation, which can jeopardize the security of countries or regions by exploiting platforms to steal personal data from millions of users. In light of such threats, employing graph analysis to locate the sources

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of cyber attacks within these networks is imperative. Identifying the initial sources of these attacks enables targeted countermeasures, enhancing the resilience of social media platforms against such threats.

Unlike node classification tasks on static topological graphs, as shown in Figure 1(a), the sample sizes of source nodes and non-source nodes often exhibit significant disparities, as shown in Figure 1(b)(c). In rumor-source identification, sources typically constitute a vanishingly small fraction of all nodes. In the extreme case only a single node originates the false information. Moreover, the diffusion dynamics are governed by the specific epidemic model of the rumor itself [41], so that (i) whether a node is infected and (ii) its intrinsic attributes represent two distinct types of evidence for the source localization tasks.

Existing methods for source localization can be categorized into three classes: (1) methods utilizing sensor observations [4, 11, 34], (2) methods having access to complete infected¹ graph [31], and (3) methods leveraging multiple infected snapshots (same graph at different timestamps) [13]. In practice, deploying sensors to gather high-fidelity observations entails significant time and resource cost [34]. Also, studies reveal that complete infected graph methods face issues from indistinguishable sources [31]. Thus, a multitude of source localization methods have opted for an approach based on learning from multiple infected snapshots [12].

Nonetheless, merely adopting an observation method based on multiple snapshots has not completely resolved the current issues. Existing multiple snapshots-based methods still encounter the following limitations.

Limitation 1: Some existing methods enhance node embeddings by enriching the information embedded. While these approaches have improved model accuracy to some extent, the highly enriched node representations may lead to a significant increase in the number of model parameters, resulting in larger and more complex models, which can be computationally expensive to train and deploy.

Limitation 2: The paucity of empirically observed misinformation cascades presents a critical bottleneck for current source localization techniques, which customarily fix a seed node in the network and generate large corpora of synthetic diffusion traces by repeatedly simulating the same spreading model, as is shown in 1(d). Such a paradigm inherently assumes that multiple, reproducible misinformation diffusion events can be initiated from an identical source. It is based on the assumption that is logistically unattainable for isolated rumor outbreaks in the real world. Consequently, models trained exclusively on these amplified trajectories risk compromised reliability and diminished robustness when deployed for real source identification.

Limitation 3: Existing representation learning-based source localization methods [13] encounter significant limitations in scenarios demanding real-time source identification. These methods often depend on retraining the entire model's parameters to adapt to new source localization tasks. In large-scale networks, the dynamism and complexity of data further intensify this challenge

which could diminish the model's generalization ability and reliability in practical applications.

Motivated by the recent success of prompt learning in both Natural Language Processing (NLP) and Graph Representation Learning (GRL), we propose LAPS- Lightweight privilege-Allocation Prompting framework for Source localization to address the aforementioned limitations. LAPS comprises five main components: Temporal Neighbourhood Sampling (TNS), Infection Snapshots Aggregation (ISA), Pre-Training of Frozen Backbone (PFB), Privilege-Allocation Prompt Graph Construction (PPGC), Prompt Graph Fine-Tuning (PGFT). For each candidate node, **TNS** extracts an adaptive-radius k-hop subgraph whose depth is automatically calibrated to the local average degree, guaranteeing sufficient contextual coverage while maintaining computational boundedness. Inside the sampled subgraph, **ISA** injects a data-driven source prior, runs a generative diffusion process, and compresses the resulting temporal infection sequence into a single infection-intensity vector that is concatenated to node features, yielding one fixed-size temporal graph per candidate. **PFB** pre-train a GNN encoder via graph-level source-region classification. After this stage, the backbone is frozen and never updated again, ensuring transferable and generic diffusion-aware representations.

TNS shifts the focus from node embedding to the construction of depth-aware subgraphs. **ISA** transitions from acquiring repetitive diffusion data with a fixed source to obtaining diffusion data based on the distribution of source priors. Together with **PFB**, these modules establish a graph-level information diffusion pre-training model that effectively mitigates the limitations caused by complex node embeddings and the paucity of repeated diffusion data for isolated rumors.

PPGC builds a compact prompt graph whose tokens are selectively attached to nodes whose infection intensity exceeds an interpretable threshold. Only these high-salience nodes receive prompt augmentation, preserving signal-to-noise ratio and interpretability. Finally, **PGFT** optimises solely the prompt tokens and their sparse adjacency by back-propagating the source localisation loss through the frozen backbone. This sub-percent parameter tuning, adjusting <1% of the total parameters, enables rapid adaptation to new tasks while retaining the generality of the pre-trained encoder.

PPGC enhances the traditional node similarity-based prompt graph linking by incorporating a privilege-allocation mechanism to strengthen the infection cascade. In conjunction with **PGFT**, these modules establish a prompt module specifically tailored for the downstream task of source localization, effectively alleviating the limitations of retraining model parameters for existing methods.

The contributions and novelty of this work can be summarized in four-fold:

- To the best of our knowledge, we are the first to introduce prompt learning into the source localization problem, establishing a novel lightweight paradigm that departs from traditional end-to-end retraining.
- We design a graph-level pre-training task that jointly performs adaptive target-node neighbourhood sampling and temporal snapshot aggregation, furnishing the frozen backbone with diffusion-aware and task-specific representations.

¹In our context, a node is *infected* if some information is passed over to it from other nodes through edges.

- We propose a privilege-allocation mechanism that selectively attaches prompt tokens only to high-salience nodes, mitigating prompt redundancy and offering transparent, infection-intensity-based explanations for each localization decision.
- We provide theoretical insights that justify both the pre-training objective and the prompt-based fine-tuning strategy, guaranteeing superior performance under data-scarce regimes.

2 Related Work

Within this section, we present an overview of pertinent literature in the domain of infection models, source localization and prompt learning.

2.1 Infection Models

Previous research has investigated various mathematical models for simulating the spread of infection across a network, such as Susceptible-Infected (SI) model [2], Susceptible-Infected-Recovered (SIR) model [1], heterogeneous SI model (HSI), heterogeneous SIR model, Independent Cascade (IC) model [15], Linear Threshold (LT) model [15], and so on. Within the SI model, each network node assumes either a susceptible (S) or infected (I) state, and once a node transitions to an infected state, it remains irrevocably so. At each timestamp, an infected node proceeds to infect its susceptible neighbors with a specified transmission probability denoted as α . Conversely, the IC model prescribes propagation probabilities to individual edges, initiating the infection process from a designated source node. An infected node enjoys a singular opportunity to activate its susceptible neighbors, with the probability of activation contingent upon the weight associated with the connecting edge. The LT model, on the other hand, assigns a threshold value, existing within the open interval to each node. Activation occurs if the cumulative influence exerted by its infected neighbors surpasses the established threshold. Game theory[21] and molecular motion[39] also provide support for the construction of diffusion models.

2.2 Source Localization

Source Localization is the inverse problem of information diffusion. The first work to apply deep learning to source localization is LPSI [36], which is based on the SI model. LPSI designs a label propagation method to give each node a feature, and then a GCN is used to detect multiple diffusion sources without knowing the underlying propagation model. GCNSI [5] further leverages GNN to enhance the prediction performance of LPSI. GCSSI [6] builds a multi-task learning-based encoder-decoder structure to predict the source node. Lately, SIGN [19] constructs more comprehensive features by considering an individual's dynamic features, and the MCGNN [28] method considers the edge features. Recently, the IVGD [31] method has designed a framework for the inverse of graph diffusion models. SL-VAE [20] incorporates the deep generative model to approximate the distribution of diffusion sources. TGASI is the state-of-the-art method for source localization, which is based on the multiple snapshot setting. TGASI considers the heterogeneous behavior diversity in time-varying infection scenarios, and a sequence-to-sequence framework is used to capture the temporal information.

2.3 Prompt Learning

Prompt learning, originally popularized in natural language processing, reformulates downstream tasks such as cloze or text completion problems by prepending learnable soft tokens to the input. This paradigm keeps the pre-trained model frozen and only tunes lightweight prompts. Representative works include GPT-3 [3], PET [25], P-tuning [22] and Prompt-Tuning [18] demonstrate that well-designed prompts can match or surpass full fine-tuning while updating less than 0.1% of the parameters.

As graph data are non-Euclidean and relational, a direct transplantation of verbal prompts is non-trivial. Recent studies show that the prompting philosophy remains applicable once “prompt” is re-interpreted as a learnable sub-structure or feature perturbation attached to the original graph.

The feasibility of graph prompting has been justified both theoretically and empirically. Wang et al. [7] provide the first guarantee that additive graph prompts can approximate bounded graph-transformation operator and derive generalization bounds for single-graph and mini-batch regimes. On the practical side, GraphPrompt [23] unifies node-level, edge-level and graph-level tasks into a single template that predicts whether a virtual prompt node connects to the rest of the graph, enabling one pre-trained model to serve multiple downstream applications. GPPT [29] instantiates this idea for node classification by converting each candidate label into a token node and recasting classification as prompted edge prediction. All-in-One [30, 32] meta-learns a shared prompt initialization that can be rapidly specialized to node/edge/graph objectives without negative transfer. Domain-specific scenarios also benefit: VPGNN [37] inserts a learnable subgraph prompt into a pre-trained GNN and achieves significant performance gains under label scarcity.

3 Preliminaries

We mainly investigate the task of source localization, with the aim of identifying sources that propagate information, given a set of observed snapshots. Initially, we provide definitions of the network, network snapshots, source localization and prompt graph.

DEFINITION 1 (NETWORKS). *An network $G = \{\mathcal{V}, \mathcal{E}, \mathbf{A}\}$ is composed of (1) the set of nodes $\mathcal{V} = \{v_1, v_2, \dots, v_n\}$, where $|\mathcal{V}| = n$; (2) the set of edges $\mathcal{E}$, where $|\mathcal{E}| = m$, an edge $e_{i,j} = (v_i, v_j) \in \mathcal{E}$ indicates that there is a connection between node v_i and v_j ; and (3) the adjacency matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ denotes the structure information of the network, where $A_{i,j} = 1$ if there is a link between nodes v_i and v_j , otherwise $A_{i,j} = 0$.*

DEFINITION 2 (NETWORK SNAPSHOTS). *Given a network, we can obtain a collection of available graph snapshots $\mathbb{G} = \{\mathcal{G}^k\}_{k=1}^T$, a snapshot graph $\mathcal{G}^k = \{\mathcal{V}, \mathcal{E}, \mathbf{Y}^k\}$ is collected on timestamp k , where $\mathcal{V}$ and $\mathcal{E}$ are the node set and edge set, and $\mathbf{Y}^k = \{y_1^k, \dots, y_{|\mathcal{V}|}^k\}$ is the infection status of nodes, and $y_i^k = 1$ if the observed state of node v_i is infected, otherwise $y_i^k = 0$.*

DEFINITION 3 (SOURCE LOCALIZATION). *The source localization problem is to find the source node $X = \{0, 1\}^{|\mathcal{V}|}$ based on the network G and observed snapshots $\mathbb{G}$, where 1 means is the original source and otherwise 0.*

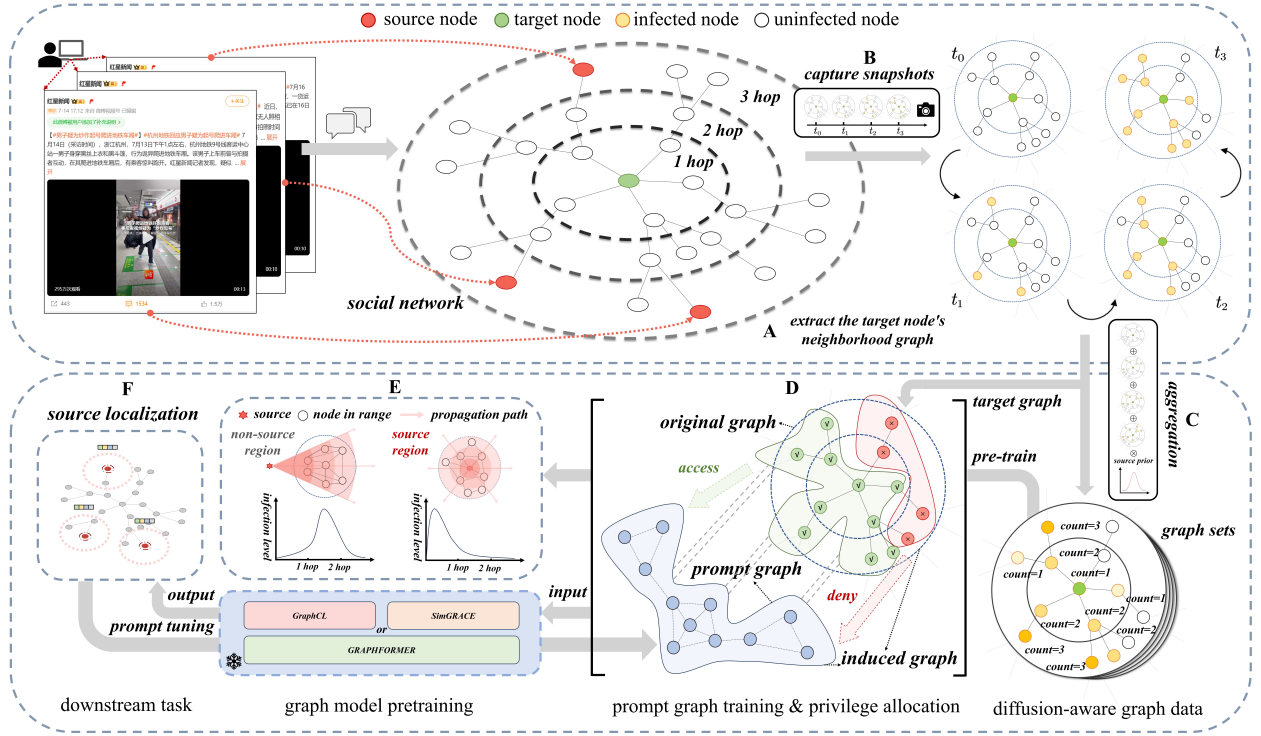


Figure 2: overview of proposed LAPS. (A) Extract the target node's neighborhood graph. (B) Capture time-resolved snapshots of the neighborhood graph. (C) Perform temporal aggregation over the captured snapshots. (D) Construct a prompt graph module and conduct privilege-allocation over the nodes of the aggregation graph. (E) The pre-training model of LAPS. (F) Conduct prompt tuning on the prompt graph by leveraging downstream task performance.

DEFINITION 4 (PROMPT GRAPH). A prompt graph for a pre-trained GNN is a learnable subgraph $G_p = (P, S)$, where $P = \{p_1, \dots, p_{|P|}\}$ is the set of prompt tokens, each token $p_i \in \mathbb{R}^d$ has the same dimension as the node features of the original graph G with $|P| \ll |V|$, and $S \subseteq P \times P$ specifies the pairwise token structure. After inserting G_p into G through an inserting pattern $\psi(\cdot, \cdot)$, the composed graph $\psi(G, G_p)$ aligns the downstream task with the pre-training pretext without updating the frozen backbone.

4 Methodology

LAPS first extracts the target node's k -hop neighbourhood, captures time-resolved snapshots of the emerging subgraph, and aggregates them into a single temporal graph. It then trains a light-weight prompt-graph module and allocates fine-grained privileges to the aggregated nodes, after which the induced graph is fed into a pre-trained backbone and the prompt graph is updated by back-propagating the source localization loss. We provide the pseudo-code of LAPS in Algorithm 1.

4.1 Pre-training for Diffusion Dynamics

To effectively model and predict diffusion processes in complex networks, it is essential to capture both the spatial and temporal characteristics of the diffusion dynamics[33]. Traditional approaches often focus on static graph representations, which fail to account for the evolving nature of diffusion over time[10, 17].

In contrast, our goal is to develop a pre-training framework that not only captures these dynamics but also specifically prepares the model for downstream source localization tasks. By pre-training on a graph-level classification task, we aim to equip the model with the ability to learn robust and generalizable representations for source region and non-source region. In this section, we introduce our pre-training strategy, which includes three key components:

Temporal Neighbourhood Sampling. For each target node we sample a k -hop neighbourhood graph, as shown in Figure 2(A). Presetting a fixed hop number k inevitably produces two extremes: in sparse graphs the sampler obtains too few nodes to yield reliable statistics, whereas in dense graphs the neighbourhood expands super-linearly and floods the encoder with redundant information[9]. Anchoring the sampling radius to the average degree $\bar{d}$ automatically calibrates the neighbourhood size: it enlarges the context when $\bar{d}$ is small and constrains it when $\bar{d}$ is large, thereby guaranteeing a lower bound on information content while preserving computational boundedness. The derivation below embeds this intuition into the notation in Section 3 and yields an adaptive hop depth $k(v_t)$ that satisfies the prescribed coverage ratio.

Given the static graph $G = (V, E, A)$, we first compute the average degree

$$\bar{d} = \frac{1}{n} \sum_{i \in V} \deg(v_i) = \frac{2m}{n}, \quad (1)$$

for a target node $v_t \in \mathcal{V}$, let $\mathcal{N}_k(v_t) = \{v \in \mathcal{V} \mid \text{dist}(v_t, v) \leq k\}$ denote its k -hop neighbourhood and define the sampling coverage ratio

$$\rho(k; v_t) = \frac{|\mathcal{N}_k(v_t)|}{n}, \quad (2)$$

to guarantee sufficient yet balanced local information, we require $\rho(k; v_t) \geq \rho_{\min}$, where $\rho_{\min} \in (0, 1)$ is a preset threshold. Dense graphs exhibit super-linear expansion of $|\mathcal{N}_k(v_t)|$, while sparse graphs suffer from insufficient context. We therefore introduce a degree-aware depth upper bound

$$k_{\max} = \left\lceil \alpha \cdot \frac{\ln n}{\ln(\bar{d} + 1)} \right\rceil, \quad (3)$$

which stems from the diameter scaling $O(\ln n / \ln \bar{d})$ of random graphs. The final sampling depth is

$$k(v_t) = \min\{k^*(v_t), k_{\max}\}, \quad (4)$$

$$k^*(v_t) = \min\{k \mid \rho(k; v_t) \geq \rho_{\min}\}.$$

The induced subgraph $\mathcal{G}_t = \mathcal{G}[\mathcal{N}_{k(v_t)}(v_t)]$ thus adapts its size to $\bar{d}$ while always satisfying the coverage constraint $\rho_{\min}$, providing a stable and informative local structure for subsequent temporal snapshot encoding.

Diffusion Source Initialization. To inject realistic source-prior knowledge into the sampled neighbourhood [14, 40], we place the synthetic diffusion origin according to a categorical prior $\pi \in \Delta^{|\mathcal{V}|-1}$ estimated from historical misinformation cascades. Specifically, we draw a source node

$$s \sim \text{Cat}(\pi), \quad \pi_i = \mathbb{P}(v_i \text{ is the origin} \mid \mathcal{H}), \quad (5)$$

where $\mathcal{H}$ denotes the empirical corpus of past cascades.

Once s is selected, we propagate the infection under a *fixed-class* generative diffusion model

$$\mathbb{P}(\mathbf{Y}^{k+1} \mid \mathbf{Y}^k, s) = \mathcal{M}_\theta(\mathbf{Y}^k, s; \mathcal{G}), \quad k = 1, \dots, T-1, \quad (6)$$

where $\mathcal{M}_\theta$ is any standard diffusion process (e.g., Independent Cascade, Linear Threshold, Simple Infectious) parameterised by θ and instantiated on the underlying graph $\mathcal{G}$. This joint procedure—source placement via π followed by $\mathcal{M}_\theta$ -based spreading—yields temporally ordered snapshots $\{\mathbf{Y}^k\}_{k=1}^T$ that respect the prior distribution of real-world sources while remaining computationally tractable for large-scale pre-training.

Temporal Snapshots Aggregation. Let $\mathbb{Y} = \{\mathbf{Y}^k\}_{k=1}^T$ denote the sequence of infection snapshots generated within the sampling window, as shown in Figure 2(B)(C). We compress this temporal signal into a single vector $\xi \in \mathbb{R}^{|\mathcal{V}_t|}$ that quantifies the *infection intensity* of every node in the induced subgraph $\mathcal{G}_t$.

For node $v_i \in \mathcal{V}_t$, we count its infections across the horizon,

$$c_i = \sum_{k=1}^T y_i^k, \quad y_i^k \in \{0, 1\}. \quad (7)$$

The infection intensity is then defined as the normalised frequency

$$\xi_i = \frac{c_i}{T} \in [0, 1], \quad (8)$$

so that $\xi_i = 0$ for never-infected nodes, while larger values reflect repeated exposure. To integrate this intensity into the node feature

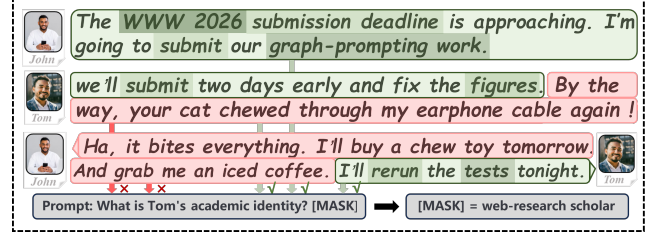


Figure 3: Incorporating the privilege-allocation mechanism into similarity-based approaches.

matrix $\mathbf{X}_t \in \mathbb{R}^{|\mathcal{V}_t| \times d}$, we append ξ as an extra column and obtain the augmented features

$$\tilde{\mathbf{X}}_t = [\mathbf{X}_t \parallel \xi] \in \mathbb{R}^{|\mathcal{V}_t| \times (d+1)}. \quad (9)$$

This single-pass aggregation preserves the temporal importance of nodes without introducing additional learnable parameters, yielding a fixed-size temporal graph $\tilde{\mathcal{G}}_t = (\mathcal{V}_t, \mathcal{E}_t, \tilde{\mathbf{X}}_t)$ ready for the subsequent pre-training phase.

Finally, the resulting graphs are fed into the designated pre-training regime to initialize the diffusion-aware backbone, as shown in Figure 2(E).

4.2 Prompt Graph Construction

Following the unified prompting paradigm proposed in *All in One* [30], we instantiate a *single, connected* prompt graph $G_p = (P, S)$ specialised for the *binary* source localization task. The token set $P = \{p_1, \dots, p_{|P|}\}$ consists of $|P| \ll |\mathcal{V}_t|$ learnable d -dimensional vectors that reside in the same feature space as the nodes of the temporal subgraph $\mathcal{G}_t$. Token-wise edges are created by the dot-product threshold pruning strategy:

$$(p_i, p_j) \in S \iff \sigma\left(\frac{p_i^\top p_j}{\|p_i\| \|p_j\|}\right) > \delta, \quad (10)$$

where $\sigma(\cdot)$ denotes the sigmoid function and δ is a fixed threshold (default 0.5). This pruning yields a *single connected component* that serves as a *global* learnable marker, eliminating the need for the multi-sub-graph design required in multi-class scenarios.

Privilege-Allocation Mechanism. In the context of misinformation propagation, the determination of a node's role as a source is critically dependent on its infection status, which is a decisive factor that cannot be overridden by other attributes, including similarity to other nodes. Traditional methods that rely solely on measuring the similarity between prompt tokens and graph nodes to dictate the inserting pattern may fail to capture this crucial aspect. In scenarios where the infection status is paramount, nodes with high similarity may not necessarily share the same infection status, thus highlighting the limitations of similarity-based approaches.

To address this shortcoming and enhance the efficiency and relevance of prompt graph insertion, we introduce a privilege-allocation mechanism that selectively connects prompt tokens to nodes in the temporal subgraph $\mathcal{G}_t$ based on their infection intensities. We illustrate the approach with an example from the NLP task, as shown in Figure 3 and Appendix B.

Let $\xi = [\xi_1, \xi_2, \dots, \xi_{|\mathcal{V}_t|}]^\top \in [0, 1]^{|\mathcal{V}_t|}$ denote the infection intensity vector as defined in Eq. (4). We define a privilege indicator function for each node $v_i \in \mathcal{V}_t$ as:

$$\delta_i = \mathbb{I}(\xi_i \geq \tau_p), \quad (11)$$

where $\tau_p \in (0, 1)$ is a preset privilege threshold, and $\mathbb{I}(\cdot)$ is the indicator function. Only nodes with $\delta_i = 1$ are eligible to receive prompt token connections.

We further define the privileged node set as:

$$\mathcal{V}_t^* = \{v_i \in \mathcal{V}_t \mid \delta_i = 1\}. \quad (12)$$

For each prompt token $\mathbf{p}_j \in \mathbb{R}^d$, we compute its relevance score with respect to each privileged node $v_i \in \mathcal{V}_t^*$ as:

$$w_{ij} = \sigma(\mathbf{p}_j^\top \mathbf{x}_i), \quad (13)$$

where $\sigma(\cdot)$ is the sigmoid function, and $\mathbf{x}_i \in \mathbb{R}^d$ is the feature vector of node v_i . The final attachment is determined by selecting the node with the highest score:

$$j^*(i) = \arg \max_j w_{ij}, \quad \text{subject to } v_i \in \mathcal{V}_t^*. \quad (14)$$

Thus, the prompt-enhanced feature vector for node v_i is updated as:

$$\tilde{\mathbf{x}}_i = \begin{cases} \mathbf{x}_i + \mathbf{p}_{j^*(i)}, & \text{if } \delta_i = 1, \\ \mathbf{x}_i, & \text{otherwise.} \end{cases} \quad (15)$$

This mechanism, as shown in Figure 2(D), ensures that only nodes with sufficient diffusion relevance are augmented with prompt information, thereby improving the signal-to-noise ratio in the input graph representation while maintaining computational efficiency and structural interpretability.

4.3 Parameter-Efficient Fine-Tuning Strategy

After the prompt graph G_p is attached to the temporal induced subgraph $\tilde{\mathcal{G}}_t$ through the privilege-allocation mechanism, the composed graph

$$\tilde{\mathcal{G}}_t^+ = \psi(\tilde{\mathcal{G}}_t, G_p), \quad (16)$$

is fed into the frozen pre-trained GNN backbone Φ_π to produce a graph-level representation

$$\mathbf{z}_t = \Phi_\pi(\tilde{\mathcal{G}}_t^+) \in \mathbb{R}^{d_z}. \quad (17)$$

A lightweight task head

$$\hat{y}_t = \text{MLP}_\phi(\mathbf{z}_t), \quad (18)$$

yields the source-region probability used for the graph-classification loss

$$\mathcal{L}_{\text{cls}} = \text{BCE}(\hat{y}_t, y_t). \quad (19)$$

Prompt tuning is performed only on the token set P and the internal token structure S of G_p , as is shown in Figure 2(F); all original-graph parameters remain frozen. Crucially, the privilege mask $\delta = [\delta_i]_{i \in \mathcal{V}_t}$ is **not** updated—nodes keep their eligibility status $\delta_i = \mathbb{I}(\xi_i \geq \tau_p)$ fixed throughout training—whereas the concrete attachment $j^*(i)$ is re-evaluated in every iteration via

$$w_{ij}^{(\ell)} = \sigma(\mathbf{p}_j^{(\ell)\top} \mathbf{x}_i), \quad j^*(i)^{(\ell)} = \arg \max_j w_{ij}^{(\ell)}, \forall i \in \mathcal{V}_t^*, \quad (20)$$

with ℓ denoting the fine-tuning step. Thus, who is privileged stays constant, but which prompt token talks to whom is allowed to evolve. The overall procedure alternates between:

Table 1: Statistics of five real-world datasets.

	Network	#Nodes $ N $	#Edges $ E $	Avg. Deg.
G_1	Karate[24]	34	78	4.6
G_2	Cora[16]	2,708	5,429	4.01
G_3	Facebook[24]	4,039	88,234	43.69
G_4	Pubmed[16]	19,717	88,648	8.99
G_5	Reddit[8]	232,965	23,213,838	199.27

- (1) forward pass through Φ_π to obtain $\hat{y}_t$;
 - (2) back-propagation of $\mathcal{L}_{\text{cls}}$ only into $\{\mathbf{p}_j\}_{j=1}^{|P|}$ and the adjacency parameters of S ;
 - (3) re-computation of attachment scores w_{ij} and update of $j^*(i)$.
- Because the trainable parameters are confined to

$$\Theta = \{\mathbf{p}_j\}_{j=1}^{|P|} \cup \{s_{jj'}\}_{(j,j') \in S}, \quad |\Theta| \ll |\Phi_\pi|. \quad (21)$$

The method achieves parameter-efficient fine-tuning with **sub-1%** tunable weights while still enabling iterative refinement of the prompt–graph interface. The process is repeated until the validation loss saturates, and the final source prediction is extracted from the converged $\hat{y}_t$ across all candidate regions.

5 Experiments

In this section, we utilize real-world datasets to assess the performance of LAPS on source localization tasks, aiming to address the following questions:

- **Q1. Effectiveness & Efficiency:** How does LAPS perform on source localization tasks compared with previous methods in terms of effectiveness and efficiency?
- **Q2. Ablation:** How does each component of LAPS contribute to the final performance?
- **Q3. Parameters Sensitivity:** How does the performance of LAPS change with different hyper-parameters?
- **Q4. Prompt-Graph Guidance:** How does LAPS enhance the representation of the original infection graph through the prompt graph?

5.1 Datasets

We conducted experiments on five real-world datasets² to evaluate the performance of localization methods, and the statistics of the datasets are provided in Table 1.

5.2 Main Performance (RQ1)

The experimental results are summarized in Table 2. The highest performance of the methods is highlighted in **boldface**, and the second-highest performance is underlined. We have the following **Observations**:

Obs 1: Under the few-shot source localization setting, LAPS attains the highest performance on all datasets in terms of Acc, F1, and AED. In the few-shot source localization scenario where no more than ten repeated historical cascades per node are available, most supervised methods underperform our proposed LAPS prompting approach. Supervised frameworks rely on multiple, node-specific

²Openly available at <http://snap.stanford.edu>.

Table 2: Source Localization Performance (%) across Accuracy (Acc $\uparrow$), F1-score (F1 $\uparrow$), and Average error distance (AED $\downarrow$).

Methods	Karate			Cora			Facebook			Pubmed			Reddit		
	Acc	F1	AED	Acc	F1	AED	Acc	F1	AED	Acc	F1	AED	Acc	F1	AED
<i>Statistic</i>															
Degree Centrality	42.5	46.2	1.25	52.1	50.4	1.15	38.0	36.9	2.70	56.2	55.1	3.45	49.3	48.0	4.05
Betweenness	49.2	47.4	1.10	54.0	52.8	1.18	36.9	35.5	2.75	58.7	57.0	3.35	50.1	49.0	3.95
Eigenvector	45.6	44.3	1.19	50.8	49.1	1.20	34.2	33.3	2.80	55.0	53.2	3.55	47.8	46.5	4.10
Closeness	31.9	33.0	1.32	48.7	46.9	1.25	33.5	32.1	2.85	54.3	52.5	3.60	47.0	45.3	4.15
Harmonic	26.2	24.9	1.18	49.8	48.2	1.18	35.0	33.6	2.78	53.9	52.0	3.58	48.2	46.8	4.08
Subgraph Coverage	29.1	28.0	1.05	55.2	54.1	1.52	39.5	38.6	2.65	60.1	59.0	3.25	51.5	50.3	3.85
Random Sampling	22.0	40.3	1.15	45.5	44.1	1.38	31.2	30.0	2.82	50.7	49.5	3.65	44.3	42.8	4.21
<i>Supervised</i>															
GCNSI [5]	11.7	13.2	1.78	5.0	6.8	1.93	4.3	1.5	2.21	0	0	2.41	0	0	2.23
GCSSI [6]	21.3	22.8	1.58	7.1	8.9	1.85	5.7	2.0	2.17	0	0	2.38	0	0	2.18
SIGN [19]	42.1	43.5	1.09	37.7	39.2	1.21	6.5	7.8	1.88	0	0	2.11	0	0	1.91
ResGCN [26]	41.0	42.8	1.11	37.1	38.5	1.17	11.7	13.0	1.69	2.5	3.8	1.89	13.0	14.5	1.61
MCGNN [28]	31.2	32.5	1.36	29.7	31.0	1.42	11.0	12.5	1.87	0	0	1.98	0	0	1.88
IVGD [31]	<u>53.7</u>	<u>55.0</u>	0.91	51.7	53.0	0.93	37.1	38.5	1.17	25.7	27.0	1.44	37.5	38.8	1.29
SL_VAE [20]	37.7	39.0	1.18	35.3	36.8	1.22	29.1	30.5	1.39	19.6	21.0	1.73	27.7	29.0	1.43
TGASI [13]	51.4	52.8	<u>0.89</u>	<u>59.6</u>	<u>60.8</u>	<u>0.72</u>	<u>52.7</u>	<u>51.0</u>	<u>1.08</u>	<u>66.1</u>	<u>67.5</u>	<u>1.42</u>	<u>67.2</u>	<u>68.5</u>	<u>1.24</u>
<i>Prompt (ours)</i>															
LAPS	63.5	62.0	0.56	64.9	64.6	0.65	55.5	54.0	1.02	70.2	68.6	1.35	69.2	69.6	1.18

diffusion annotations that are costly or impossible to obtain in reality; in contrast, LAPS first pre-trains on large-scale simulations driven by the source-prior diffusion, alleviating the demand for repeated annotations. The subsequent privilege-allocation prompt further boosts few-shot performance. Compared with supervised baselines, LAPS improves accuracy by 2.8%–9.8% and F1 by 1.1%–7.0%, while reducing AED by 0.06–0.33 across all datasets.

Obs 2: LAPS is also lightweight and efficient compared with competing methods. As listed in Table 3, we select the two large-scale datasets—Pubmed and Reddit—to report the number of trainable parameters and the total training time for IVGD, SL_VAE, TGASI and LAPS. Owing to the limited number of prompt tokens, the tunable parameter space of LAPS is far smaller than that of traditional methods; meanwhile, by eliminating complex node-level embedded recomputation, the training speed is significantly boosted.

Table 3: Comparison of parameter volume and running time across methods for Pubmed and Reddit datasets.

Methods	Pubmed		Reddit	
	Parameter	Time (s)	Parameter	Time (s)
IVGD	~ 61K	~ 8.5×10^2	~ 75K	~ 1.5×10^3
SL_VAE	~ 43K	~ 1.2×10^3	~ 52K	~ 2.5×10^3
TGASI	~ 232K	~ 1.1×10^3	~ 292K	~ 2.4×10^3
LAPS	~ 2.8K	~ 1.2×10^1	~ 3K	~ 9.9×10^1

5.3 Ablation Study (RQ2)

In this section, we investigate the effects of various components of LAPS on the performance of the source localization task. We

Table 4: Ablation study (%) of LAPS on Cora, Facebook, and Pubmed datasets F1-score (F1 $\uparrow$), and Average error distance (AED $\downarrow$).

Methods	Cora		Facebook		Pubmed	
	F1	AED	F1	AED	F1	AED
LAPS	64.6	0.65	54.0	1.02	68.6	1.35
w/o PA	53.2	0.95	38.7	1.68	49.1	1.89
w/o PG	55.8	0.88	40.5	1.55	51.3	1.75
w/o Diffusion	12.4	1.58	16.6	2.11	11.2	2.32
w/o SDP	58.1	0.78	48.9	1.12	62.0	1.38
w/o TNS	60.3	0.71	50.8	1.08	64.5	1.37

classify these components into two groups and perform an ablation study on each group. The results are compiled in Table 4.

5.3.1 Pre-training component. In this group, we ablate the three core designs that equip the pre-training model with diffusion-aware representations, consisting of the Temporal Neighbourhood Sampling(**w/o TNS**), Source-Distribution Prior(**w/o SDP**) and the Diffusion (**w/o Diffusion**). To remove the TNS, We disable the degree-aware hop-depth adaptation by fixing $k = 2$ for every target node. This leads to sub-graphs that are either uninformative or computationally prohibitive. To remove SDP, we replace the data-driven source prior π with selecting the source node uniformly at random. Without the historical-bias injection the generated cascades lose their realism: the model seldom encounters hub-like seeds that dominate real-world diffusion traces, so the learnt representation systematically underestimates the centrality footprint of true origins. To remove diffusion, we remove the synthetic infection

phase. the subgraph is fed into the encoder with zero-infection features $\xi = 0$. As Table 4 shows, F1 collapses to 11%–17% and AED jumps beyond 2.0, confirming that explicit diffusion dynamics is the dominant factor of SL-relevant signal.

5.3.2 Prompt component. In this group we ablate the two core designs that realise task-specific adaptation, consisting of Privilege-Allocation (w/o PA) and the Prompt Graph (w/o PG) itself. To remove PA, we disable the infection-intensity mask, allowing *every* node in the sampled subgraph to receive a prompt-token augmentation. Consequently, the model injects learnable parameters into vertices that are either uninfected or only marginally activated, thereby diluting the gradient signal and increasing computational overhead. To remove PG, we discard the entire prompt graph G_p and feeds the frozen GNN with the plain temporal subgraph $\tilde{G}_t$. Without the learnable global tokens the model loses its task-specific inductive bias and collapses to a generic graph-classifier. Overall, the prompt-centric ablations corroborate that (i) privilege-allocation filters out noisy gradient paths and (ii) the prompt graph itself acts as a lightweight yet powerful task adapter, jointly contributing to the superior performance of LAPS.

5.4 Parameters Sensitivity (RQ3)

In this section, we further analyze the sensitivity of the proposed framework LAPS on Karate. We have identified four key parameters to evaluate the sensitivity. τ_p determines node eligibility for prompt attachment based on infection intensity; δ is the cosine threshold for creating token-wise edges; $\rho_{\min}$ specifies the minimum coverage ratio for neighborhood sampling; and P denotes the number of tokens in the prompt graph. Figure 4 illustrates the impact of these parameters on the performance of LAPS.

By systematically adjusting each parameter while keeping others constant, we can isolate the specific effects of each on the model's output. The model's performance exhibits a generally convex trend with respect to τ_p and δ , with optimal values found near 0.4 and 0.6, respectively. In contrast, performance with respect to $\rho_{\min}$ and P tends to increase, suggesting a positive correlation. To maintain the model's lightweight nature, we select these superior values early on as the model parameters.

5.5 Prompt Graph Guidance (RQ4)

In this section, we aim to demonstrate the contribution of the prompt to the graph infection representation. To verify this, we fixed two nodes with lower degrees, v_1 and v_2 , as two constant sources. On the Karate dataset, we conducted diffusions based on IC, LT and SI with infection rates $I = 0.2, 0.4, 0.6, 0.8$. We mapped the original graph and the prompt-induced graph corresponding to these two sources into two sets of one-dimensional vectors and calculated the cosine similarity for the rest. This approach measures the contribution of the prompt graph to the graph infection representation. As shown in Figure 5, the introduction of the prompt graph increases the difference in representations of diffusion graphs generated by different source nodes. Moreover, as the infection rate increases, the improvement effect of the prompt graph becomes increasingly significant.

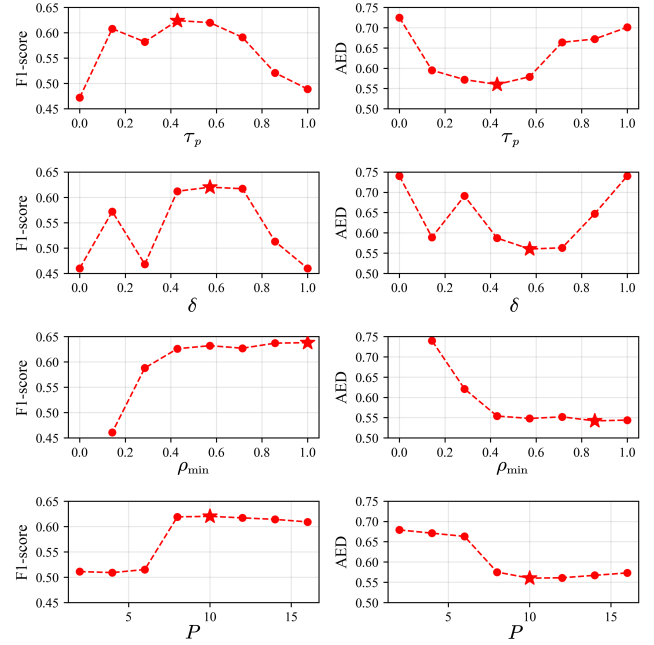


Figure 4: Performance with different parameters values.

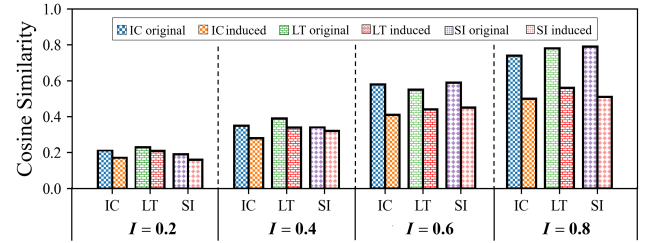


Figure 5: Performance with different parameters values.

6 Conclusion

In this paper, we investigate the few-shot source localization problem in social networks. Building on recent advances in NLP and graph prompting, we propose a novel prompting framework that integrates graph-level source-region pre-training with a privilege-allocation mechanism. Extensive experiments demonstrate the effectiveness of our framework.

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A Pseudo-code of LAPS

Algorithm 1 Pseudo-code of LAPS in a PyTorch-like style.

```

import torch
from torch_geometric.data import Data, Batch

def laps_target_sample(G, vt, d_bar, rho_min, alpha):
    # Adaptive k-hop sampling for target node vt
    k_max = int(alpha * log(G.num_nodes) / log(d_bar + 1)) + 1
    for k in range(1, k_max + 1):
        subg = k_hop_subgraph(vt, k, G)
        if subg.num_nodes / G.num_nodes >= rho_min:
            break
    return subg # returns subgraph object

def infection_intensity(Y):
    # Y: [T, N] -> xi: [N]
    return Y.float().mean(dim=0) # xi_i = c_i / T

def privilege_mask(xi, tau_p=0.4):
    # delta: [N] 1 if xi_i >= tau_p
    return (xi >= tau_p).float()

def prompt_attach(x, P, delta, cos_thresh=0.6):
    # attach prompt tokens to privileged nodes
    # P: [P, d] learnable tokens
    # x: [N, d] node features
    # delta: [N] privilege indicator
    V_star = delta.nonzero(as_tuple=False).flatten()
    if V_star.numel() == 0:
        return x
    x_star = x[V_star] # [V*, d]
    scores = torch.sigmoid(x_star @ P.t()) # [V*, P]
    j_star = scores.argmax(dim=1) # [V*]
    x[V_star] += P[j_star] # residual update
    return x

for epoch in range(max_epochs):
    for subg in loader: # subg: sampled temporal graph
        xi = infection_intensity(subg.Y) # [N]
        delta = privilege_mask(xi, tau_p=0.4)
        subg.x = prompt_attach(subg.x, P, delta, cos_thresh=0.6)
        with torch.no_grad():
            z = backbone(subg) # frozen SimGRACE/GCN
        y_hat = mlp(z) # source-region logits
        loss = BCEWithLogitsLoss(y_hat, subg.y)
        loss.backward() # update P & MLP only
        optimizer.step()
        optimizer.zero_grad()

```

B DIAGRAM OF PROMPTING IN LAPS

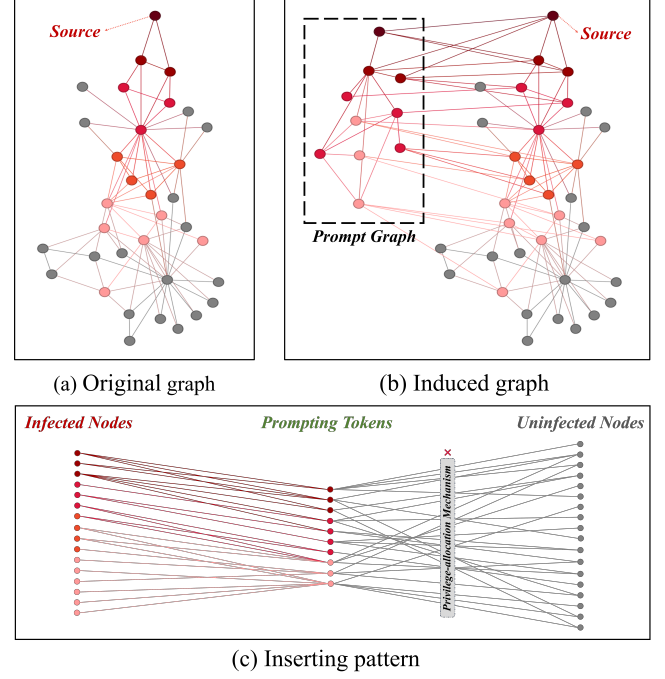


Figure 6: Diagram of privilege-allocation prompting.

C SELECTION OF MODELS

Table 5: Selection of pre-training and fine-tuning models for Cora and Facebook.

Methods	Cora			Facebook		
	Acc	F1	AED	Acc	F1	AED
GraphCL+GAT	64.8	64.5	0.70	54.5	53.8	1.04
GraphCL+GCN	63.9	64.2	0.77	54.0	53.4	1.05
GraphCL+GT	63.8	64.1	0.71	54.9	53.2	1.11
SimGRACE+GAT	64.5	64.0	0.68	54.3	53.9	1.08
SimGRACE+GCN	64.9	64.6	0.65	55.5	54.0	1.02
SimGRACE+GT	64.2	64.1	0.86	55.0	53.5	1.03

Table 6: Selection of pre-training and fine-tuning models for Pubmed and Reddit.

Methods	Pubmed			Reddit		
	Acc	F1	AED	Acc	F1	AED
GraphCL+GAT	69.2	67.5	1.39	69.0	68.8	1.21
GraphCL+GCN	69.7	67.8	1.45	68.8	68.6	1.25
GraphCL+GT	70.1	68.0	1.38	68.5	69.2	1.23
SimGRACE+GAT	68.8	68.2	1.37	68.5	68.9	1.27
SimGRACE+GCN	70.2	68.6	1.35	69.2	69.6	1.18
SimGRACE+GT	69.8	68.1	1.37	68.8	69.2	1.24

D EXPERIMENTAL SETTINGS

Comparison methods. We illustrate the performance of LAPS on various experiments against previous state-of-the-art methods, including seven statistic methods and eight supervised methods.

Implementation details. For each sampled source node we repeat the cascade generation ten times under the same source distribution; every cascade is observed for $T = 5$ steps and infection intensities ξ_i are averaged across the ten runs. We adopt SimGRACE[38] for pre-training, and a lightweight GCN[16] for prompt tuning while keeping the SimGRACE encoder frozen (Appendix C). The prompt graph contains $|P| = 10$ learnable tokens, pruned at cosine threshold $\delta = 0.6$. Nodes with $\xi_i \geq \tau_p = 0.4$ are eligible for token attachment. GCN prompt tuning uses Adam at 5×10^{-4} with cosine decay for 150 epochs. Diffusion is simulated by the Independent Cascade (IC) model with infection rate $I = 0.3$. The source prior π is frequency-estimated from training cascades. Monte-Carlo roll-outs are executed on the full graph $\mathcal{G}$: for each sampled subgraph $\mathcal{G}_t$ we independently draw 1 000 diffusion realisations by (i) sampling a

source node from π and (ii) running the IC model on $\mathcal{G}$ for $T = 5$ steps, node states inside $\mathcal{G}_t$ are then extracted to form temporal snapshots.

Evaluation metrics. We employ three metrics to assess the performance of LAPS: 1). Accuracy (Acc): Given that source localization amounts to a binary classification problem, the F1-score is widely employed as an evaluation metric. 2). F1-Score (F1): Given that source localization amounts to a binary classification problem, the F1-score is widely employed as an evaluation metric. 3). Average Error Distance (AED): AED is a metric used to quantify the minimum distance between the predicted source node and the ground-truth source node. The definition of AED is as follows:

$$\Delta = \min_{P \in \text{permutation}(R)} \sum_{i=1}^K \frac{d(r_i^*, p_i)}{K}, \quad (22)$$

where $K = |R^*|$ is the number of sources. $r_i^* \in R^*$ is the index of ground truth sources, R is the set of predicted sources and $P = \{p_1, p_2, \dots, p_K\}$ is the single permutation of R .